The Quine-McCluskey (Tabular) Method

1. Why Quine-McCluskey?

Theory: Karnaugh Maps are excellent visual tools, but they become highly impractical and prone to human error when you move beyond 4 or 5 variables. The Quine-McCluskey (Q-M) method is an algorithmic, tabular alternative that solves these issues.

Feature	Karnaugh Map	Quine-McCluskey
Max Variables	4 (Good), 5-6 (Very Hard)	Unlimited
Format	Visual / Easy to draw	Tabular
Computer Implementable?	No	Yes (ESPRESSO Algorithm)
Risk of Error	High (Easy to miss overlapping groups)	Low (Follows strict rules)
Finds Minimum Cover?	Depends on user skill	Yes (Exhaustive search)

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2. Crucial Terminology

Before starting the algorithm, you must understand these two definitions:

- **Prime Implicant (PI):** An implicant (grouping) that CANNOT be combined with any other term to eliminate another variable. In a K-Map, this is a rectangle made as large as possible. In Q-M, these are the terms that never get a checkmark (✓) next to them.
- **Essential Prime Implicant (EPI):** A Prime Implicant that is the *only* one covering a specific minterm. It **must** be included in your final Boolean expression.

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3. The 8-Step Algorithm

Here is the exact step-by-step process for minimizing an expression using Q-M:

Phase 1: Finding Prime Implicants

1. **Group Minterms:** Write all minterms in binary. Sort them into groups based on the number of `1`s they contain (e.g., Group 0 has zero `1`s, Group 1 has one `1`, etc.).

2. **Build Column I:** Compare every minterm in Group k with every minterm in Group $k+1$. If they differ by exactly *one* bit, combine them. Place a dash (`-`) where the bit changed, write the new term in Column II, and mark a check (✓) next to the original terms.
3. **Build Column II, III...:** Repeat the process. Two terms can only combine if their dashes (`-`) are in the exact same position AND they differ by exactly one non-dash bit.
4. **Identify Prime Implicants:** Any term in any column that does NOT receive a checkmark (✓) is a Prime Implicant.

Phase 2: The Prime Implicant Chart
5. **Build the PI Chart:** Create a grid. The rows are your un-checked Prime Implicants. The columns are the original required minterms. Place an `X` in the grid wherever a PI covers a minterm.
6. **Find Essential PIs (EPIs):** Scan the columns. If a column has only *ONE* `X` in it, the Prime Implicant in that row is Essential. Circle it. It must be in your final answer.
7. **Select Minimum Cover:** After identifying EPIs, cross out all minterm columns covered by those EPIs. If there are still uncovered minterms, select the fewest number of remaining non-essential PIs needed to cover them.
8. **Write Final SOP:** Convert the selected PIs from binary/dash notation back into Boolean variables ($0 = \text{bar}$, $1 = \text{normal}$, dash = eliminated).

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4. Don't Cares in Quine-McCluskey

Theory:

Handling "Don't Cares" (indicated by d or X) in Q-M follows a very strict two-part rule:

1. **In Phase 1 (Building Columns):** Treat Don't Cares exactly like regular required minterms. Include them in your groups to help generate larger Prime Implicants.
2. **In Phase 2 (Building the PI Chart): Omit** the Don't Cares from the top columns of your chart! You do not want to force your final equation to cover a minterm that you don't actually care about.

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5. Visualizing

We are going to minimize this function:

$$f(a,b,c,d) = \Sigma m(0,1,2,5,6,7,8,9,10,14)$$

Step 1: Column I (Group by the number of 1s)

First, convert every minterm into its 4-bit binary form. Then, organize them into groups based on how many `1`s are in the binary number. (Group 0 has zero 1s, Group 1 has one 1, etc.).

Group	m	a	b	c	d	
Group 0	0	0	0	0	0	✓
Group 1	1	0	0	0	1	✓
	2	0	0	1	0	✓
	8	1	0	0	0	✓
Group 2	5	0	1	0	1	✓
	6	0	1	1	0	✓
	9	1	0	0	1	✓
	10	1	0	1	0	✓
Group 3	7	0	1	1	1	✓
	14	1	1	1	0	✓

(Note: I have put a ✓ next to every number because every single one of these ends up combining in the next step.)

Step 2: Column II (Combine Pairs)

Now, compare every number in Group 0 with every number in Group 1. If they differ by **exactly one bit**, combine them and put a dash (`-`) where the bit changed.

Then compare Group 1 to Group 2, and Group 2 to Group 3.

(For example, comparing m_0 `0000` and m_1 `0001` gives us `000-`).

Group	Pairs	a	b	c	d	
Group 0	$(0,1)$	0	0	0	–	✓
	$(0,2)$	0	0	–	0	✓
	$(0,8)$	–	0	0	0	✓
Group 1	$(1,5)$	0	–	0	1	★ (Prime Implicant!)
	$(1,9)$	–	0	0	1	✓
	$(2,6)$	0	–	1	0	✓
	$(2,10)$	–	0	1	0	✓
	$(8,9)$	1	0	0	–	✓
	$(8,10)$	1	0	–	0	✓
Group 2	$(5,7)$	0	1	–	1	★ (Prime Implicant!)
	$(6,7)$	0	1	1	–	★ (Prime Implicant!)
	$(6,14)$	–	1	1	0	✓
	$(10,14)$	1	–	1	0	✓

Notice the three terms with stars (★). They do not match with anything in the next step. Because they cannot be simplified any further, they are officially **Prime Implicants**.

Step 3: Column III (Combine Quads)

Now, repeat the exact same process for Column II. You can only combine two pairs if their dashes (`-`) are in the **exact same spot** and they differ by exactly one other bit.

(For example, comparing $(0,1)$ `000-` with $(8,9)$ `100-` gives us `-00-`).

Group	Quads	a	b	c	d	
Group 0	$(0,1,8,9)$	–	0	0	–	★ (Prime Implicant!)
	$(0,2,8,10)$	–	0	–	0	★ (Prime Implicant!)
Group 1	$(2,6,10,14)$	–	–	1	0	★ (Prime Implicant!)

No further combinations can be made. Every item here gets a star.

Step 4: Converting PIs back to Algebra

Let's list all the starred terms from Step 2 and Step 3, converting the 0s and 1s back into letters ($0 = \text{bar}$, $1 = \text{normal}$).

1. $(0,1,8,9)$ forms `-00-` $\rightarrow \bar{b}\bar{c}$
2. $(0,2,8,10)$ forms `-0-0` $\rightarrow \bar{b}\bar{d}$

3. $(2, 6, 10, 14)$ forms `- - 1 0` $\rightarrow \mathbf{c\bar{d}}$
4. $(1, 5)$ forms `0 - 0 1` $\rightarrow \mathbf{\bar{a}\bar{c}d}$
5. $(5, 7)$ forms `0 1 - 1` $\rightarrow \mathbf{\bar{a}bd}$
6. $(6, 7)$ forms `0 1 1 -` $\rightarrow \mathbf{\bar{a}bc}$

Step 5: The Prime Implicant Chart

Now we build a grid. The rows are our 6 Prime Implicants. The columns are the 10 original minterms from the problem. We place an `X` wherever the PI covers that specific number.

Prime Implicants		0	1	2	5	6	7	8	9	10	14
$(0, 1, 8, 9)$	$\mathbf{\bar{b}\bar{c}}$	X	X					X	**X**		
$(0, 2, 8, 10)$	$\bar{b}\bar{d}$	X		X				X		X	
$(2, 6, 10, 14)$	$\mathbf{c\bar{d}}$			X		X				X	**X**
$(1, 5)$	$\bar{a}\bar{c}d$		X		X						
$(5, 7)$	$a'bd$				X		X				
$(6, 7)$	$a'bc$					X	X				

Finding Essential Prime Implicants (EPIs):

Look down the columns. Are there any columns with only a single `X` in them?

- Look at Column **9**. The only X is in row $(0, 1, 8, 9)$.
- Look at Column **14**. The only X is in row $(2, 6, 10, 14)$.

Because they are the *only* groups that cover those numbers, they are **Essential**. They must be in your final answer.

Current Equation: $F = \bar{b}\bar{c} + c\bar{d} \ldots$

Step 6: Selecting the Minimum Cover

Because we selected $(0, 1, 8, 9)$ and $(2, 6, 10, 14)$, we have successfully covered the numbers: `0, 1, 8, 9` and `2, 6, 10, 14`.

Look at the original list of numbers: `0, 1, 2, 5, 6, 7, 8, 9, 10, 14`.

The only numbers left that are NOT covered yet are **5** and **7**.

We need to pick one more Prime Implicant to cover both 5 and 7. Look back at our list of PIs:

- PI $(1, 5)$ covers 5.
- PI $(6, 7)$ covers 7.
- PI $(5, 7)$ covers **both** 5 and 7!

To keep the equation as small as possible, we select $(5, 7)$, which is $\bar{a}bd$.

The Final Minimized Equation

By adding our two Essential PIs and our final covering PI, we get the exact answer from the slides:

$$f(a, b, c, d) = \bar{b}\bar{c} + c\bar{d} + \bar{a}bd$$